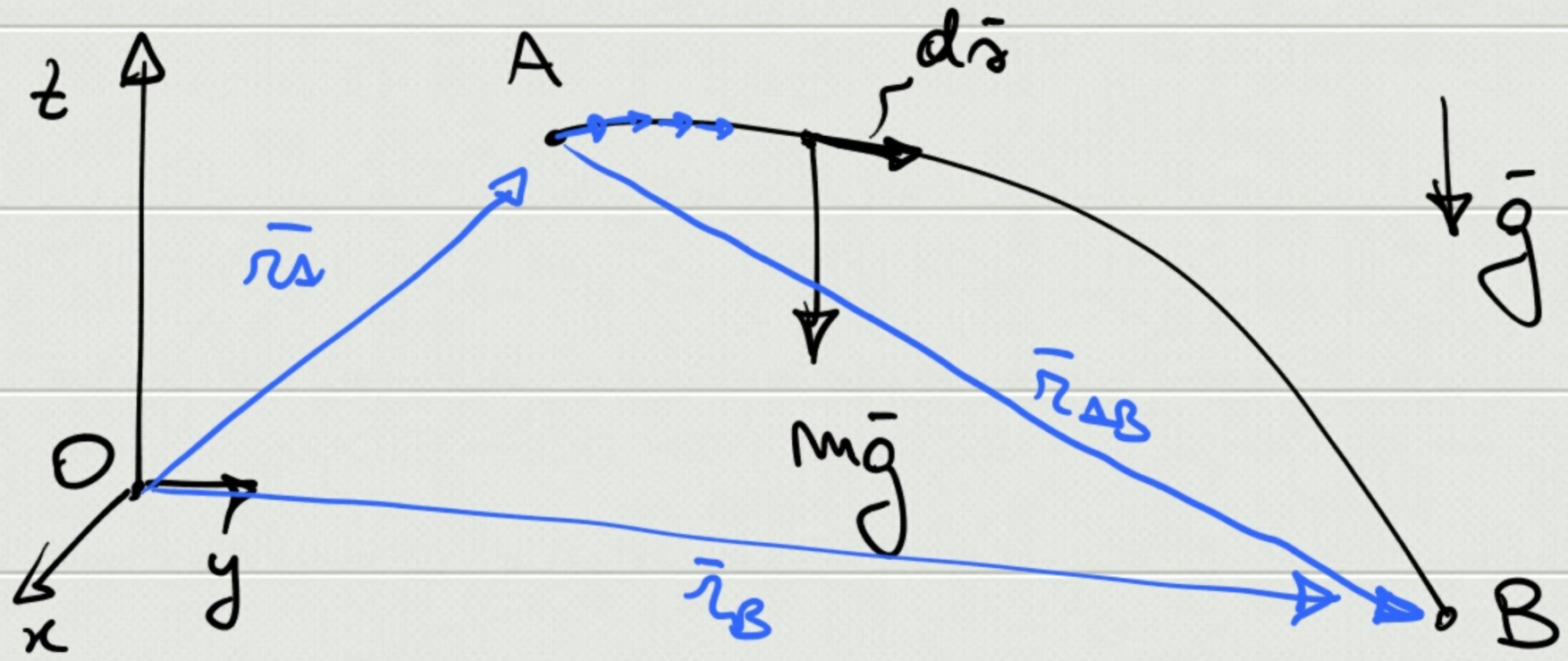
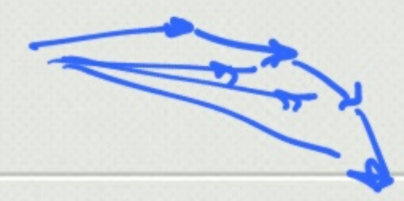


$$W_{A\to B} = \int_A^B m\bar{g}\, d\bar{s} =$$



$$= m\bar{g}\int_A^B d\bar{s} = m\bar{g}\cdot\bar{r}_{AB} = m\bar{g}\,(\bar{r}_B - \bar{r}_A) = *$$

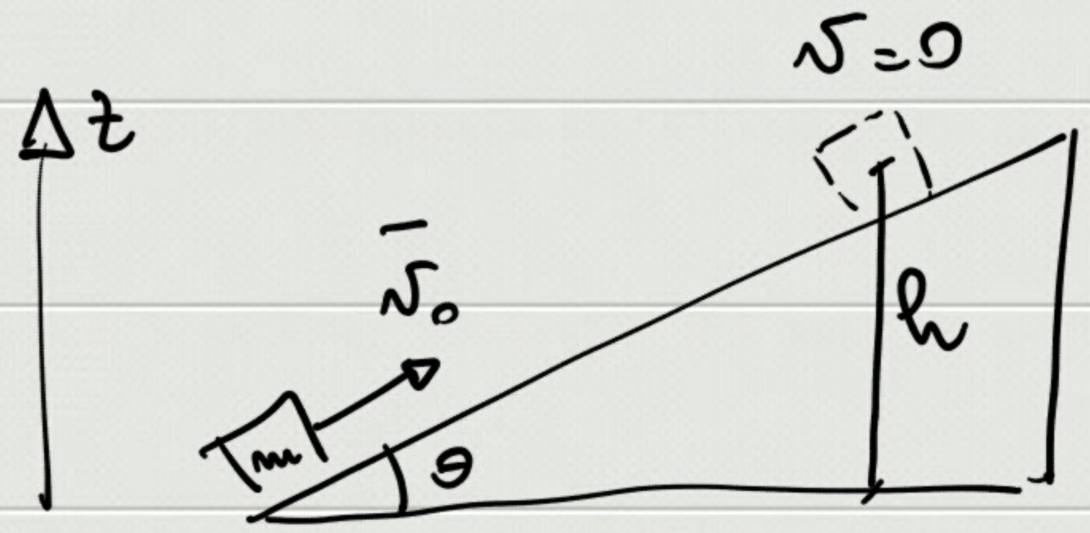
$$\bar{r} = x\,\bar{u}_x + y\,\bar{u}_y + z\,\bar{u}_z \qquad \bar{g} = -g\,\bar{u}_z \quad (g>0)$$

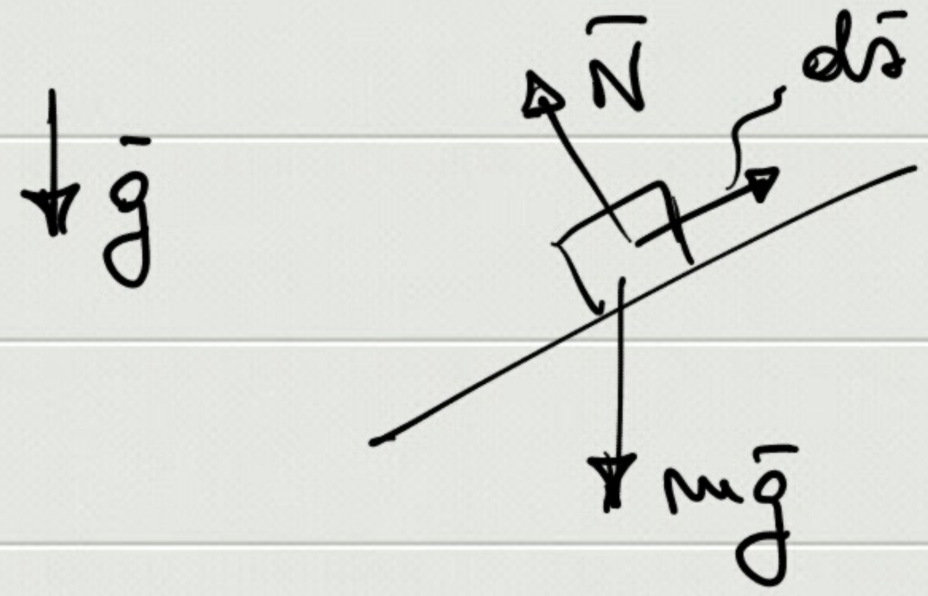
$$* = -mg\,(z_B - z_A)$$

Energia potenziale : $\boxed{E_p}$

$$\boxed{E_{p,peso} = mgz}$$

$$\Rightarrow W_{A\to B} = -\left(E_{p,peso,B} - E_{p,peso,A}\right) = -\Delta E_{p,peso}$$





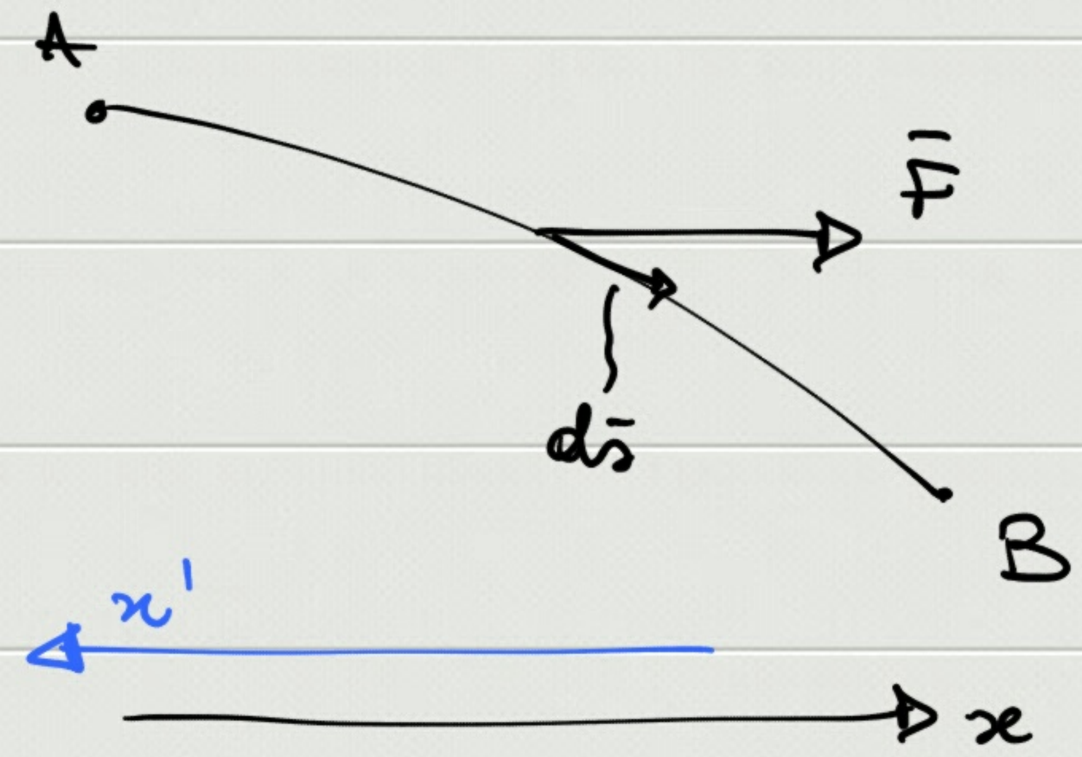
$$W_{TOT,\, i \to f} = W_{peso,\, i \to f} = -\Delta E_{p,\, peso} = -\left(mg\, z_f - mg\, z_i\right) =$$

$$= -mgh$$

$$W_{TOT,\, i \to f} = \Delta E_k = \cancel{\frac{1}{2} m v_f^2}^{\,=0} - \frac{1}{2} m v_i^2 = -\frac{1}{2} m v_0^2$$

$$\Rightarrow \boxed{h = \frac{v_0^2}{2g}}$$

$\bar{F} = \text{const} = F\,\bar{u}_x$



$$W_{A\to B} = \int_A^B \bar{F}\,d\bar{s} = \int_A^B \bar{F}\cdot\left(dx\,\bar{u}_x + dy\,\bar{u}_y + dz\,\bar{u}_z\right) =$$

$$= \int_{x_A}^{x_B} F\,dx = F\,(x_B - x_A) = *$$

$$\boxed{E_{p,F} = -Fx} \qquad \boxed{E'_{p,F} = Fx'}$$

$$* = -Fx_A + Fx_B = -\left(E_{p,F,B} - E_{p,F,A}\right) = -\Delta E_{p,F}$$

$$W_{A\to B} = -\Delta E'_{p,F} = -\left(Fx'_B - Fx'_A\right)$$